DBMS Assignment 4

Airline Reservation System

1. Comparison

Aspect	Relational database	NoSQL database
Structure	Tables, rows, keys and relationships.	Documents, key-value, wide-column or graph data.
Scalability	Can scale up and also scale out with suitable systems.	Often designed for distributing data across many servers.
Consistency	Usually strong transactions and constraints.	Depends on product; may offer strong, tunable or eventual consistency.
Performance	Good for joins, reports and transactions.	Good for access patterns designed around keys or documents.
Schema	Defined schema and managed changes.	Often more flexible schema.
Use cases	Bookings, banking, orders and inventory.	Flexible content, event data, caching and large distributed workloads.

2. Real-World Examples

A relational database such as MySQL or PostgreSQL can store passengers, flights, bookings and payments with foreign keys and transactions. A document database such as MongoDB can store flexible travel guides or reviews. Redis can cache frequently requested search results. These are common examples of use, not claims about a specific company's internal systems.

3. Why Organizations Choose Different Databases

Organizations consider data structure, transaction needs, consistency, traffic, speed, cost, expertise and maintenance. Some systems use more than one database, such as a relational database for bookings and a cache for search results.

4. Recommendation

Use MySQL or PostgreSQL for the core Airline Reservation System because its records are related and seat allocation must remain consistent. Use NoSQL only for a clear additional need, such as flexible content or high-volume event data.

5. Conclusion

Neither database type is best for every task. The correct choice depends on the application's requirements and workload.